F-EMBEDDINGS OF A FINITELY GENERATED EXTENSION CORRESPOND TO COMPATIBLE ROOT TUPLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: This bijection is the input for the product formula |Mor_F(K, overline{F})| = prod deg_s(P_i) (b8 sibling card L48380).

Editorial note. The label lemma-count-embeddings names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter Situation situation-finitely-generated (source lines 1485-1496): K = F(alpha_1..alpha_n) finite, tower K_i = F(alpha_1..alpha_i), P_i the minimal polynomial of alpha_i over K_{i-1}, algebraic closure overline{F} fixed
- Universal property of quotient rings: a homomorphism out of K_{i-1}[x]/(P_i) over K_{i-1} is the same as a choice of a root of P_i^{phi_{i-1}} in the target; this is also why each step map phi_i exists and is unique
- The section paragraph immediately preceding the lemma (source lines 1499-1505): a morphism phi: K -> overline{F} yields step maps phi_i and phi(alpha_i) is a root of the conjugated polynomial P_i^{phi_{i-1}}

Source reference. Original reference 09H6: sources/stacks-fields/pinned-original-source.tex, source lines 1486–1486. Prerequisite pointer invoked at 1509; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09H6 the correspondence*

$$\mathrm{Mor}_F(K, \overline{F}) \longrightarrow \{(\beta_1, \ldots, \beta_n) \text{ as below}\}, \quad \varphi \longmapsto (\varphi(\alpha_1), \ldots, \varphi(\alpha_n))$$

is a bijection. Here the right hand side is the set of n-tuples $(\beta_1, \ldots, \beta_n)$ of elements of $\overline{F}$ having the following property:

1. *$\beta_1 \in \overline{F}$ is a root of P_1; let $\varphi_1 : K_1 \to \overline{F}$ be the homomorphism over F sending α_1 to β_1,*
2. *$\beta_2 \in \overline{F}$ is a root of $P_2^{\varphi_1}$; let $\varphi_2 : K_2 \to \overline{F}$ be the homomorphism extending φ_1 sending α_2 to β_2,*
3. *and so on until,*
4. *$\beta_n \in \overline{F}$ is a root of $P_n^{\varphi_{n-1}}$.*

In each step the homorphism φ_i exists and is unique because $K_i = K_{i-1}[x]/(P_i)$ and β_i is a root of $P_i^{\varphi_{i-1}}$.

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Proof. The map from left to right is discussed above the lemma. Let $(\beta_1, \ldots, \beta_n)$ be an element of the right hand side. Then we let $\varphi : K = K_n \to \overline{F}$ be the unique homorphism extending φ_{n-1} sending α_n to β_n. Uniqueness implies that the two constructions are mutually inverse. $\square$